

Python ML by Example
AI Stuart Russel (Pearson)

→ AI: Mimic Human Intelligence

↳ Making machines intelligent and enable to take action & decisions that humans do

→ Machine - Makes life easier

↳ Uses power to control mechanical movements

Complex machines: 2 or more simple machines

→ Intelligence: Ability to understand, learn & think

→ 2 dimensions of AI:

(1) Humans vs. Rational

(2) Act vs. Think

→ Acting Humanly Turing Test

Total Turing Test

→ (1) Think Humanly

(2) Act Rationally

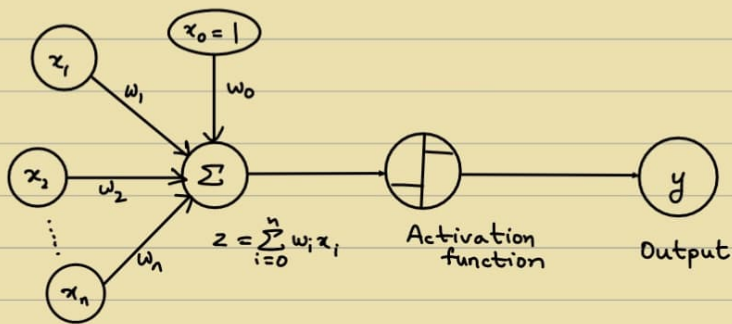
(3) Think Rationally

→ ML: Construction of Algos that can learn from & make predictions on data.

↳ features



→ Perceptron Model



$$y = f(z) = \begin{cases} 1, & \text{if } z \geq 0 \\ 0, & \text{if } z < 0 \end{cases}$$

→ Knowledge Base :

Set of Sentences represented in knowledge representation language.

Expressing knowledge explicitly in a computer tractable way
(logic)

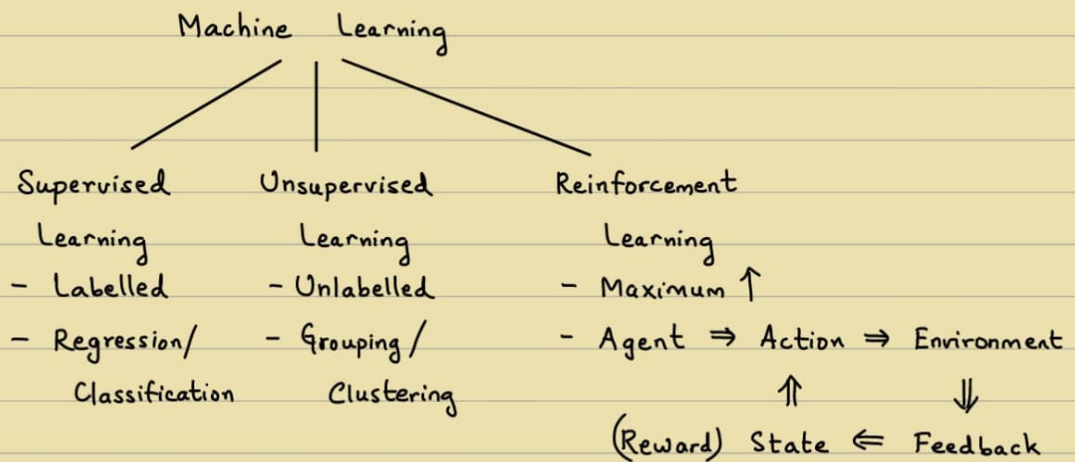
→ Logic : A way of thinking or explaining something

↳ Basis of all mathematical & Automated reasoning

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→ Machine Learning :

- Perform specific tasks effectively without explicit instructions
- Relies on Patterns and Inference instead.
- Involves building a math model based on training data (sample data) to make predictions / decisions
- Why ML :
 - Data ↑ - Various formats
 - Patterns ↑



→ AlphaGo :

MuZero → Without Rules

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Machine Learning

- Term coined around 1960
- Why learn ? Why not just hire enough programmers and code in rules ?
 - Lots of patterns for an activity/event
 - Events can be dynamic
 - **Data** is increasing exponentially
 - **Data** is also in various formats [Text, Audio, Video]
 - Higher quality **data** due to cheaper storage
- Can be broadly classified into three categories
 - Unsupervised, Supervised and Reinforcement learning

AlphaGo

- First computer Go program to defeat a 9-dan professional player
- Uses Monte Carlo Tree search algorithm based on knowledge learned by a deep learning method
- Beat World No. 1 ranked player in 2017
 - Retired after this match
- <https://deepmind.google/research/breakthroughs/alphago/>
- <https://www.youtube.com/watch?v=WXuK6gekU1Y>
- AlphaGo Zero – Version without human data and stronger than AlphaGo [defeated 100-0]

↪ Reinforcement learning
(Design incentive structure)

• Search:

Initially AI used BFS, then later switched to DFS.

- (1) BFS
 - (2) DFS
 - (3) A*
- } - Non-Randomised

(4) Monte Carlo Tree Search - Randomized Search Technique

Simulation - performed by Neural Networks (AlphaGo)

→ Which direction to go to ?

AlphaZero & MuZero

- AlphaZero, a generalized version of AlphaGo Zero. Took 4 hours to learn Chess and defeat reigning world computer chess champion 28 to 0 in 100 matches
- https://www.youtube.com/watch?time_continue=7&v=tXIM99xPQC8
- MuZero: Master games without knowing rules
- Uses approach similar to AlphaZero, developed in 2019

AlphaFold: solution to a 50 year old grand challenge in biology

- <https://deepmind.com/blog/article/alphafold-a-solution-to-a-50-year-old-grand-challenge-in-biology>
- Figuring out what shapes proteins fold into is known as the "protein folding problem" - grand challenge in biology for the past 50 years
- Focus of intensive scientific research for many years, using a variety of experimental techniques such as nuclear magnetic resonance and X-ray crystallography.

AlphaFold

- Number of ways a protein could theoretically fold before settling into its final 3D structure is astronomical.
- Cyrus Levinthal estimated 10^{300} possible conformations for a typical protein.
- Estimated would take longer than the age of universe to enumerate all possible configurations. Yet in nature, proteins fold spontaneously, some within milliseconds - referred to as Levinthal's paradox.

Nobel Prize in Chemistry 2024 !!!

- The Nobel Prize in Chemistry 2024 is about proteins, life's ingenious chemical tools.
- David Baker has succeeded with the almost impossible feat of building entirely new kinds of proteins.
- Demis Hassabis and John Jumper have developed an AI model to solve a 50-year-old problem: predicting proteins' complex structures. These discoveries hold enormous potential.

AI Index 2025

- Source: Stanford Institute for Human-Centered Artificial Intelligence (HAI)
- Report: 2025 AI Index Report
- <https://hai.stanford.edu/ai-index/2025-ai-index-report>
- Slides follow

AI Performance on Demanding Benchmarks

- AI achieved large gains on harder benchmarks
- Strong progress in coding and video generation
- AI agents outperform humans in time-constrained tasks

Industry Dominates AI Development

- ~90% of notable models built by industry
- Performance gaps narrowed
- Compute scale continues to rise - training compute doubles every five months, datasets every eight, and power use annually.

→ Cross - Validation :

Partitioning of training dataset.

k parts $\equiv$ Training : $k-1$ parts
 Testing : k^{th} part } → Repeat 'k' times
 k ↓

- Holdout Method - Non-exhaustive Cross Validation

Con : Some data never gets into the test data



- Exhaustive Cross-Validation :

LOOC - Used for Sparse dataset

- Nested Cross Validation :

k -fold Cross validation with validation and Test

Perform K -fold on the partitioned data (without Test)

→ Regularization :

Penalty function

Reduce importance for some terms

Occam's Razor : Simplest hypothesis that fits data should be preferred. (Not Complex model)

(i) Early Stopping

Given :

$$f(x) = \theta + \theta_1 x + \theta_2 x^2 + \theta_3 x^3$$

$$\text{MSE} = \frac{1}{m} \cdot \min_{\theta, \theta_1, \theta_2, \theta_3} \sum_{i=1}^m (\hat{f}_{\theta}(x^{(i)}) - y^{(i)})^2$$

$$\text{Loss} : J(\theta)$$

Adding Penalty : $1000 \theta_3 + 1000 \theta_4$

(ii) Feature Selection

Pick top k Heuristically useful features

n features : 2^n feature combinations

Filter Methods : Correlation

(Chi-Squared for Categorical
 Pearson coeff. for Continuous)

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→ Feature Selection : (Reduce overfitting)

2^n total

(i) Filter Methods :

x , $2x$ - High correlation, $\therefore$ Remove Redundancy

(ii) Wrapped methods :

Underfit Model Search

Recursive feature elimination : Remove best feature and construct the next model iteratively (Worst)

(iii) Embedded methods :

Assess features acc. to Accuracy

e.g. Regularisation - Add Penalty in loss function

• Dimensionality Reduction :

- Information loss : Minimum

• Data Preprocessing :

(Data Mining)

- CRISP DM - Standard

- Data preparation $\Rightarrow$ Modeling $\Rightarrow$ Evaluation $\Rightarrow$ Deployment

- Data Cleaning - ★

Domain knowledge - ★

- Manual work (Numerical)

- Features can be quantitative / Categorical.

- Feature Engg. : Improving features

- Missing values

→ Ignore

→ Substitute with fixed values : Imputing

- Label Encoder : Label $\Rightarrow$ Numeric value

(Integer Label Encoder) - ML infers ordering based on encoding.

Done by One-hot encoding

- Scaling :

(i) Standardization : $\mu = 0$ & $\sigma^2 = 1$ $\left(\frac{z - \mu}{\sigma} \right)$

If not Gaussian, Subtract Median & Divide by Interquartile
1st (25%) and 3rd (75%) Range

(ii) Normalization : min(0) - max(1) Scaling

- Polynomial features :

Occam's Razor : Avoid Higher order & interaction terms
 $a^2 + ab + b^2$, ab : interaction term

$$y = w_1x_1 + w_2x_2 + w_3$$

- Power transformations :

- Box-Cox transform : Find best power
- Binning : Separation of feature values into bins
(Lossy transformation)

→ Uncertainty - Decision Theory

Uncertainty

- May not have categorical answers
 - May not know exactly which route is slow
 - 50% chance that one route is slow
 - Even if we know it is slow, how slow?
- **Uncertainty** changes the way an agent makes a decision
- Probability theory used in helping agents reason with uncertainty
 - Summarize uncertainty due to our ignorance or laziness
 - May be very very costly to remove such uncertainty
- **Rational decision** depends on
 - Relative importance of various goals
 - Likelihood or degree to which they will be achieved

Decisions under Uncertainty

- How should an agent make a decision:
 - Plan 1, 80% chance it will succeed (Route 1)
 - Route 1 will be quick if flowing smoothly, but stuck for 1 hour if accident
 - If plan1 is successful the outcome is very good, but failure is very bad
 - Plan 2, 70% chance it will succeed (Route 2)
 - Route 2 will be somewhat quick, but not so bad even if slow
 - If plan2 successful the outcome is good (not as good as plan1's success), but failure is not that bad

Basic Probability Notation

- Suppose we have probabilities over all possible values of a variable -- denote by $P(X)$
 - Called the probability distribution for the random variable:
 - $P(\text{Weather} = \text{sunny}) = 0.6$
 - $P(\text{Weather} = \text{rain}) = 0.1$
 - $P(\text{Weather} = \text{cloudy}) = 0.29$
 - $P(\text{Weather} = \text{snow}) = .01$
 - $P(\text{Weather}) = \langle 0.6, 0.1, 0.29, 0.01 \rangle$
- $$\sum_{x_i \in \{\text{sunny, rain, cloudy, snow}\}} P(X=x_i) = 1$$

Diagnosis domain: Introducing Cavity Variable

- Doctor diagnosing a dental patient's toothache
- Propositional logic rules
- Toothache $\rightarrow$ Cavity [Toothache implies Cavity] (Incomplete rule)
- Toothache $\rightarrow$ Cavity $\vee$ Gum Problem $\vee$ Abscess ... (Lots of such problems)
- Turning the rule around, Cavity $\rightarrow$ Toothache
 - Not totally right either, need to make it logically exhaustive

Inference using full joint distribution

	toothache		$\neg$ toothache	
	catch	$\neg$ catch	catch	$\neg$ catch
cavity	.108	.012	.072	.008
$\neg$ cavity	.016	.064	.144	.576

- 3 variables : {Cavity, Toothache, Catch}
- Dentist probe catches or not
- $2 * 2 * 2$ table
- $P(\text{Cavity}) = .108 + .012 + .072 + .008 = 0.2$ (Called **Marginal Probability** of Cavity)

Marginalization (or summing out)

- For the variable set Z not including variable Y

$$P(Y) = \sum_{z \in Z} P(Y, z) = \sum_z P(Y | z) P(z)$$

$$P(\text{Cavity}) = \sum_{z \in \{\text{Catch, Toothache}\}} P(\text{Cavity}, z)$$
- The above rule is called conditioning (conditioned on z)
- **Marginalization** and **conditioning** very useful rules

Marginalization

$$P(\text{cavity} | \text{toothache}) = \frac{P(\text{cavity} \wedge \text{toothache})}{P(\text{toothache})}$$

$$= \frac{0.108 + 0.012}{(0.108 + 0.012 + 0.016 + 0.064)} = 0.6$$

$$P(\neg \text{cavity} | \text{toothache}) = \frac{P(\neg \text{cavity} \wedge \text{toothache})}{P(\text{toothache})}$$

$$= \frac{0.016 + 0.064}{0.108 + 0.012 + 0.016 + 0.064} = .4$$

$$P(\text{cavity} | \text{toothache}) + P(\neg \text{cavity} | \text{toothache}) = 1$$

Marginalization

- Denominator can be viewed as a normalization constant $1/P(\text{toothache})$
- $P(\text{Cavity} | \text{Toothache}) = \alpha P(\text{Cavity, toothache}) = \alpha [P(\text{Cavity, toothache, catch}) + P(\text{Cavity, toothache, } \neg \text{catch})] = .108 + .012 = \alpha 0.12$
- $\alpha P(\neg \text{Cavity, toothache}) = \alpha [.016 + .064] = \alpha 0.08$
- Normalizing $\alpha \langle .12, .08 \rangle$ gives $\langle .6, .4 \rangle$
- **Issue:** For a domain with n Boolean variables, input table size of $O(2^n)$
- Full joint distribution in tabular form not a practical tool for building reasoning systems

Marginalization

- Denominator can be viewed as a normalization constant $1/P(\text{toothache})$
- $P(\text{Cavity} | \text{Toothache}) = \alpha P(\text{Cavity}, \text{toothache}) = \alpha [P(\text{Cavity}, \text{toothache}, \text{catch}) + P(\text{Cavity}, \text{toothache}, \neg \text{catch})] = .108 + .012 = \alpha 0.12$
- $\alpha P(\neg \text{Cavity}, \text{toothache}) = \alpha [.016 + .064] = \alpha 0.08$
- Normalizing $\alpha <.12, .08>$ gives $<.6, .4>$
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Independence

- Helps in reducing size of domain representation and complexity of inference problem
- Let's add weather into the example : sunny, rain, cloudy, snow $\rightarrow 2 * 2 * 2 * 4$ entries in table
- $P(\text{toothache}, \text{catch}, \text{cavity}, \text{cloudy}) = P(\text{cloudy} | \text{toothache}, \text{catch}, \text{cavity}) P(\text{toothache}, \text{catch}, \text{cavity})$
- $P(\text{cloudy} | \text{toothache}, \text{catch}, \text{cavity}) = P(\text{cloudy})$

Bayes' Rule

- Given that
 - $P(A \wedge B) = P(A|B) * P(B)$
 - $P(A \wedge B) = P(B|A) * P(A)$
 - $P(B|A) = (P(A|B) * P(B)) / P(A)$
- We can determine $P(B|A)$ given $P(A|B)$, $P(B)$ and $P(A)$
- $P(\text{effect} | \text{cause})$ may be **causal knowledge**, $P(\text{cause} | \text{effect})$ **diagnostic knowledge**
- Often have conditional probabilities on causal relationships $\rightarrow P(\text{symptoms} | \text{disease})$
 - Need to infer diagnostic knowledge many times

Bayes' Rule Example

- S: Proposition that patient has stiff neck
- M: Proposition that patient has meningitis
- Meningitis causes stiff-neck, 70% of the time
- Given:
 - $P(S | M) = 0.7$
 - $P(M) = 1/50,000$
 - $P(S) = .01$
 - $P(M|S) = P(S | M) * P(M) / P(S) = (.7 * 1/50000) / .01 = .0014$

Bayes' Rule Use

- Why not estimate $P(M | S)$ the same way as we do $P(S|M)$?
 - Why do we need Bayes' rule?
- $P(S | M)$ is **causal knowledge**, does not change
 - It is "model based"
 - It reflects the way meningitis works
- $P(M | S)$ is **diagnostic**; tells us likelihood of M given symptom S
 - Diagnostic knowledge may change with circumstance, thus helpful to derive it
 - If there is an epidemic, probability of Meningitis goes up; rather than trying to estimate $P(M | S)$ which is very hard, we can compute it using the other terms
 - However, $P(S|M)$ is unaffected by the epidemic

Computing the denominator: $P(S)$

We wish to avoid computing the denominator in the Bayes' rule

- May be hard to obtain
- In our example, the denominator is $P(S)$
 - Techniques to compute (or avoid computing $P(S)$)

Our first approach is to compute relative likelihoods

- If M (meningitis) and W (whiplash) are two possible explanations:
 - $P(M|S) = P(S | M) * P(M) / P(S)$
 - $P(W|S) = P(S | W) * P(W) / P(S)$
- $P(M|S) / P(W|S) = P(S | M) * P(M) / P(S | W) * P(W)$

- Another approach: Approach #2
 - $P(M|S) = P(S | M) * P(M) / P(S)$
 - $P(\text{NOT}(M)|S) = P(S | \text{NOT}(M)) * P(\text{NOT}(M)) / P(S)$
- $P(M|S) + P(\text{NOT}(M)|S) = 1$ (These two quantities must sum to 1)
- $[P(S | M) * P(M) / P(S)] + [P(S | \text{NOT}(M)) * P(\text{NOT}(M)) / P(S)] = 1$
- $P(S | M) * P(M) + P(S | \text{NOT}(M)) * P(\text{NOT}(M)) = P(S)$
- Calculate $P(S)$ in this way...

Conditional Independence

- Toothache and Catch are independent given the presence of Cavity
 - Each is directly caused by cavity, toothache is caused by cavity and a dentist catches cavity using probe
 - A toothache cannot be caught by a probe nor a probe results in a toothache
- Conditional independence captured as : $P(\text{toothache} \wedge \text{catch} | \text{cavity}) = P(\text{toothache} | \text{catch}, \text{cavity}) P(\text{catch} | \text{cavity})$
- $P(\text{toothache} | \text{catch}, \text{cavity}) = P(A | B, C)$
 - If A and C are conditionally independent given B
 - Then, probability of A is not dependent on C
 - i.e. $P(A | B \wedge C) = P(A | B)$
- Therefore $P(\text{toothache}, \text{catch} | \text{cavity}) = P(\text{toothache} | \text{cavity}) P(\text{catch} | \text{cavity})$

More General Forms of Bayes' Rule

- $P(Y | X) = P(X | Y) * P(Y) / P(X)$
- Bayes' rule for multi-valued variables
- Generalize to some background evidence e
- $P(Y | X, e) = P(X | Y, e) * P(Y | e) / P(X | e)$

Combining Evidence

- Revisiting the Meningitis example --
- S: Proposition that patient has stiff neck
- H: Proposition that patient has severe headache
- M: Proposition that patient has meningitis
- Meningitis causes stiff-neck, 50% of the time
- Meningitis causes head-ache, 70% of the time
- Probability of Meningitis should go up, if both symptoms reported -- how to combine such symptoms?

Combining Evidence

- $P(A | B \wedge C) = P(A \wedge B \wedge C) / P(B \wedge C)$
- Numerator:
 - $P(M \wedge S \wedge H) = P(S | M \wedge H) * P(M \wedge H)$
 $= P(S | M) * P(M \wedge H) \rightarrow \text{WHY?}$
 $= P(S | M) * P(H | M) * P(M)$

Conditional Independence

- Helps decompose larger tables into small ones
 - Allows probabilistic systems to scale up
- **Conditional independence** much more common than absolute independence
- Decomposition of large probabilistic systems into weakly connected subsets through conditional independence
 - Considered one of the most important developments in the recent history of AI
- Will deal more in lectures on Belief nets